

Tejashwi K J

Robotics & Automotive Engineer · MSc Robotics, Högskolan Väst · Trollhättan, Sweden
tejashwijagadish@gmail.com · [Github](#) · tejashwi-jagadish.github.io

PROFILE

Robotics & Automotive Engineer with 3+ years of industry experience at Tier-1 OEM suppliers (KPIT Technologies, Bosch). MSc graduate from Högskolan Väst specialising in industrial automation, intelligent systems, and safety-critical design. Combines hands-on vehicle electrical harness engineering with academic depth in PLC programming (TwinCAT 3, IEC 61131-3), control theory, and embedded systems. Experienced across mechanical, electrical, and software domains — with a track record of delivering production-ready solutions. Open to Robotics, Automation, or Automotive EE research and engineering roles.

TECHNICAL SKILLS

Engineering & CAD	Solid Edge Cabling, CATIA V5 Electrical, AutoCAD, Design Validation, Harness Routing
Industrial Automation	TwinCAT 3, PLC Programming (FBD, ST, SFC, IEC 61131-3), RAPID (ABB Robotics), HMI Design
Automotive EE	Vehicle Electrical Harness Design, 3D Routing, Connector Placement, Teamcenter PLM
Control Systems	Control Theory, P Controller Tuning, SFC Sequencing, Fail-Safe Architecture
Safety & Validation	Safety-Critical Design, Functional Validation, Regulatory Compliance, QA Processes
Programming	Python, RAPID, IEC 61131-3 Structured Text, Excel Automation, Git
Soft Skills	Cross-functional Collaboration, Technical Documentation, Manufacturing Readiness

ENGINEERING PROJECTS

AGV HMI Control System | University West 2026

TwinCAT 3 · IEC 61131-3 (ST, FBD) · HMI Design

- Designed a full operator control system for a Virtual AGV using TwinCAT 3 with dual-instance (PLCSim + PLCStudent) architecture and inter-instance communication.
- Implemented stop-latch and reset-to-HOME phase recovery logic, plus a reusable FB_AGVComm function block for encapsulated AGV communication.
- Built a custom HMI displaying Moving/Ready/Stop indicators and live Position output; all logic in IEC 61131-3 Structured Text.

AI Powered Visual Defect Detection on EV battery Packs | University West 2026 — Ongoing

Deep Learning.Computer Vision. Python

- Building a real time defect detection pipeline to identify visual and localise surface defects on EV battery cells and modules from camera imagery.
- Applying deep learning object detection.
- Targets production line speed and scalability for high volume EV manufacturing

- Designed advanced front monoshock suspension for motorcycle applications, covering structural integrity, material selection, and dynamic characterisation.
- Performed FEA for stress distribution and fatigue life prediction; optimised geometry for handling stability, ride comfort, and manufacturing feasibility.

WORK EXPERIENCE

- Design Engineer | KPIT Technologies

Oct 2022 – Aug 2025

 - Engineered vehicle electrical harnesses with comprehensive 2D drawings and 3D routing across multiple OEM production programmes using Solid Edge Cabling and CATIA V5.
 - Executed strategic connector placement and manufacturability validation, ensuring compliance with international automotive electrical standards.
 - Provided production launch support including installation guidance and technical documentation managed through Teamcenter PLM.
 - Collaborated with cross-functional teams to optimise harness designs for cost, weight, and assembly efficiency; reviewed designs against client specs.
- Engineering Intern | Bosch

May 2022 – Jul 2023

 - Managed electrical harness data architecture using Solid Edge and Excel for complex automotive programmes; prepared compliance and manufacturing documentation.
 - Supported quality assurance and design review cycles; created detailed technical reports for cross-functional engineering teams.

EDUCATION

- MSc in Robotics | Högskolan Väst, Trollhättan, Sweden

2025 – Ongoing

Specialisation in industrial automation and intelligent systems. Coursework: RAPID (ABB) programming, PLC systems, control theory, machine learning in robotics, safety-critical system design.
- BE in Mechanical Engineering | Dayananda Sagar University, Bangalore, India

2018 – 2022

Foundation in mechanical design, thermodynamics, manufacturing processes, CAD/CAM, and structural analysis. Final-year project: Front Monoshock Suspension Design with FEA validation.

TOOLS & PLATFORMS

Expert	Solid Edge Cabling, CATIA V5, Teamcenter PLM, Technical Documentation, Design Validation
Advanced	TwinCAT 3, Python, RAPID (ABB), AutoCAD, IEC 61131-3 PLC, Control Systems.
Languages	English (Professional) · Swedish (Conversational)

PERSONAL QUALITIES

Systems Thinker · Safety-First · Production Ready · Detail-Oriented · Collaborative. Passionate about engineering at the intersection of mechanical, electrical, and software disciplines — with a drive to build systems that move, sense, and decide reliably.